LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	24
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need for simple oxygen system and pressurized cabin effect of lack of o2, Types of 02 regulators
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need for simple oxygen system and pressurized cabin effect of lack of o2, Types of 02 regulators. Fire detection and protection system:
5. Specific Objectives	17.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of regulator operation principles; Safety: Do not breathe directly from regulator; Instructor supervision required
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft oxygen system?
(b) How does cabin pressure affect oxygen availability at altitude?
(c) Name one type of oxygen regulator used in aviation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Physiological effects of hypoxia • Need for simple oxygen systems • Pressurized cabin oxygen requirements • Types of oxygen systems • Working principles of components	L & Ds	PPT, IB	15 min
(b)	• Types of 02 regulators • Demand regulator operation • Continuous flow regulator operation • Pressure regulator function • Regulator maintenance basics	L & Ds	PPT, PDF	20 min
(c)	• Comparison of regulator types • Selection criteria for systems • Safety protocols for oxygen use • Troubleshooting common regulator issues • Integration with fire protection systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why a simple oxygen system is necessary for emergency descent.
(b) Differentiate between demand and continuous flow regulators based on operation.
(c) Describe the effect of lack of oxygen on crew performance at high altitude.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from LP 24 to LP 02, focusing on how fire detection requirements relate to oxygen system safety.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Types of fire zone classification of a/c, detection system requirements, types of detection system, types of detectors, extinguishing agents and systems	Refer to the master syllabus reference list	LP 25

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min